

Abdul Basit Khattak

 [abdulbasit.github.io](https://github.com/abdulbasit)

 github.com/abdulbasit

 [linkedin.com/in/basit](https://www.linkedin.com/in/basit)

 abdulbasit@lums.edu.pk

EDUCATION

Lahore University of Management Sciences (LUMS)

09/22 – 05/26

Bachelor's degree, Computer Science (Advised by Imdadullah Khan)

- Completed nine teaching assistantships during undergraduate studies
- Relevant Coursework: Algorithms, Applied Probability, Calculus II, Data Structures, Deep Learning, Generative AI, Linear Algebra, Network Security, Principles and Techniques of Data Science, Software Engineering, Statistics & Data Analysis

Islamia College Peshawar

09/20 – 05/22

High School (Subjects: Maths, Phy & Chem)

Score: 95.45/100

- Ranked 2nd in graduating batch of 2022
- Best in Mathematics across the board

PROJECTS

Decarbonizing Pakistan: MESSAGEix Analysis [\[Notebook\]](#) [\[Paper\]](#)

10/25 – 12/25

- Modeled Pakistan's energy system using MESSAGEix to evaluate low-emission pathways for 2025–2070
- Assessed a cumulative CO₂ emissions cap, carbon pricing, renewable energy integration, carbon capture & storage (CCS), and investment requirements under stringent climate targets
- *Technologies: MESSAGEix, GAMS, Python, Optimization Modeling, Data Analysis*

Engagement Signals at Scale: Predicting YouTube Trending Speed [\[Notebook\]](#)

10/25 – 11/25

- Analyzed 2017/2018 US/UK/CA YouTube trending data and engineered timing, text-length, & category features
- Ran exploratory data analysis on engagement, country, and category effects; built and compared regression/ensemble models to predict time-to-trend, evaluated with MSE/MAE/R², and saved artifacts for reproducible inference
- *Technologies: Python, pandas, scikit-learn, seaborn/matplotlib, Statistical Testing*

SALER x SQUID GAME [\[Demo\]](#)

08/25

- Designed a collection of 3 strategic board games inspired by Netflix's *Squid Game* featuring dynamic theming & responsive design
- Built a custom in-app music player with persistent cross-page playback and playlist controls
- *Technologies: JavaScript, CSS, HTML, GSAP, Web Audio API*

ColorGrid [\[Demo\]](#)

08/25

- Developed a real-time multiplayer 5×5 grid game with profile customization, persistent game history, coin rewards, and global leaderboards
- Built server-side matchmaking, robust turn validation, real-time game-state synchronization, and a flood-fill algorithm for calculating connected-area scores
- *Technologies: React, Node.js, Express.js, Socket.IO, MongoDB, Tailwind CSS, JWT*

EXPERIENCE

Lahore University of Management Sciences

Lahore, Pakistan

Teaching Fellow

06/26 – Present

- MATH230: Probability (Summer 2026)

Undergraduate Teaching Assistant

09/23 – 05/26

- CS360: Software Engineering (Spring 2026)
- DISC325: Business Data Management (Spring 2026)
- CS310/CS5102: Algorithms (Fall 2025)
- CS501/EE515/MATH439: Applied Probability (Fall 2025)
- MATH230: Probability (Spring 2025)
- MATH102: Calculus II (Fall 2024)
- MATH120: Linear Algebra with Differential Equations (Summer 2024)
- MATH101: Calculus I (Summer 2025), (Fall 2023)

TECHNICAL SKILLS

Programming Languages: C++, C, Java, JavaScript, Python **ML/AI:** NumPy, Pandas, Matplotlib, Scikit

Web/App Development: HTML, CSS, React, Next.js, Node.js

Miscellaneous: Git, MongoDB, Data Analysis, Mathematical Modeling, L^AT_EX